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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

*Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT303/20BT303 (17BT302) – UNIT OPERATIONS IN BIOPROCESS INDUSTRIES(UNIT OPERATIONS)

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Distinguish between the following:				
	i) Real and ideal fluids	5	L*1	1	1
	ii) Newtonian and Non-Newtonian fluids	5	L3	1	2
	iii) Compressible and incompressible fluids				
	b) Derive Newton's Law of viscosity.				
	c) Derive the equation to measure pressure at a point using a simple U-tube manometer.	10	L3	1	2
2.	a) Outline Reynold's apparatus and describe the importance of Reynold's experiment in fluid mechanics.	10	L2	2	2
	b) Derive velocity equation for flow of viscous fluids through circular pipes.	10	L3	2	2
3.	a) Prove that acceleration is a derived unit.	5	L2	1	1
	b) What is dimensional analysis? What is its importance? List the methods of dimensional analysis.	5	L2	1	1
	c) Derive the energy balance equation.	10	L3	2	2
		Unit – II			
4.	a) Describe filtration operation and any one type of filter.	10	L2	3	1
	b) Derive the expression for terminal settling velocity.	10	L3	3	2
5.	a) Summarize the different laws for size reduction.	10	L2	4	2
	b) Describe how sieve analysis is used in the determination of average particle size.	10	L2	4	2
6.	a) Stating the assumptions of Kynch theory, outline the process of thickener design.	10	L3	3	2
	b) Explain about propeller and paddle agitators used in bioprocess operations.	10	L2	4	2
		Unit – III			
7.	a) Explain the working principle of reciprocating pump.	10	L2	5	2
	b) Derive the flow equation for venturi meter.	10	L2	5	2
8.	a) Describe the construction and calibration of rotameter.	10	L2	5	2
	b) What is the application of centrifugal pump in bioprocess operation? Explain its design.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
